Practice set 3 - Solutions

1. Compute each of the following expression if it is defined. If an expression is undefined, explain why.

   Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}$.

   (a) $2A - 3B$.

   (b) CA.

   (c) $(AB^t)C$.

   Answer.

   (a)
   $$2A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}, 3B = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 3 & -6 \end{bmatrix}, 2A - 3B = \begin{bmatrix} -1 & 4 & 3 \\ 8 & 7 & 18 \end{bmatrix}.$$

   (b)
   $$CA = \begin{bmatrix} (-1)(1) + (1)(4) & (-1)(2) + (1)(5) & (-1)(3) + (1)(6) \\ (2)(1) + (3)(4) & (2)(2) + (3)(5) & (2)(3) + (3)(6) \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 14 & 19 & 24 \end{bmatrix}.$$

   (c)
   $$B^t = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -2 \end{bmatrix}, AB^t = \begin{bmatrix} 4 & -4 \\ 10 & -7 \end{bmatrix}, (AB^t)C = \begin{bmatrix} 4 & -4 \\ 10 & -7 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} -12 & -8 \\ -24 & -11 \end{bmatrix}.$$

   $\square$

2. Let
   $$A = \begin{bmatrix} 1 & 1 & 3 \\ z & 7 & x-2 \\ 3 & y & 3 \end{bmatrix}, \ B = \begin{bmatrix} 1 & 1 & 3 \\ z & 7 & 3x \\ 3 & y+2 & 3 \end{bmatrix}.$$

   Suppose $A = B^t$, find x, y and z.

   Answer.
   $$\begin{bmatrix} 1 & 1 & 3 \\ z & 7 & x-2 \\ 3 & y & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ z & 7 & 3x \\ 3 & y+2 & 3 \end{bmatrix}^t = \begin{bmatrix} 1 & z & 3 \\ 1 & 7 & y+2 \\ 3 & 3x & 3 \end{bmatrix}$$
   $$\Rightarrow \begin{cases} x - 2 = y + 2 \\ z = 1 \\ y = 3x \end{cases} \Rightarrow \begin{cases} x = -2 \\ y = -6 \\ z = 1 \end{cases}$$

   $\square$

3. Let A be a 5×7 matrix given by

$$[A]_{i,j} = 2i + j, \quad 1 \le i \le 5, 1 \le j \le 7,$$

and B be a be a 7×6 matrix given by

$$[B]_{i,j} = (-1)^i, \quad 1 \le i \le 7, 1 \le j \le 6.$$

(a) Find $[AB]_{2,3}$. Show your steps clearly.

(b) Try to find $[AB]_{i,k}$ for each pair (i, k), where $1 \le i \le 5$ and $1 \le k \le 6$.

Answer.

(a)

$$\begin{aligned}
[AB]_{2,3} &= [A]_{2,1}[B]_{1,3} + [A]_{2,2}[B]_{2,3} + [A]_{2,3}[B]_{3,3} + [A]_{2,4}[B]_{4,3} \\
&\quad + [A]_{2,5}[B]_{5,3} + [A]_{2,6}[B]_{6,3} + [A]_{2,7}[B]_{7,3} \\
&= (5)(-1) + (6)(1) + (7)(-1) + (8)(1) + (9)(-1) + (10)(1) + (11)(-1) = -8.
\end{aligned}$$

(b)

$$\begin{aligned}
[AB]_{i,k} &= \sum_{j=1}^{7} [A]_{i,j}[B]_{j,k} \\
&= \sum_{j=1}^{7} (2i + j)(-1)^j \\
&= 2i \sum_{j=1}^{7} (-1)^j + \sum_{j=1}^{7} j(-1)^j \\
&= 2i(-1 + 1 - 1 + 1 - 1 + 1 - 1) + (-1 + 2 - 3 + 4 - 5 + 6 - 7) \\
&= -2i - 4.
\end{aligned}$$

$\square$

4. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad D = \operatorname{diag}(1, -1) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Suppose A and DA are both symmetric. Show that A is diagonal.

Answer. Since A is symmetric, we have $A = A^t$, which means

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

By comparing the entries, we find that the above equality is equivalent to $b = c$. Therefore,

$$A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}.$$

Next, consider

$$DA = \begin{bmatrix} a & b \\ -b & -d \end{bmatrix}.$$

If DA is symmetric, we have $DA = (DA)^t$, which implies

$$\begin{bmatrix} a & b \\ -b & -d \end{bmatrix} = \begin{bmatrix} a & -b \\ b & -d \end{bmatrix}.$$

By comparing the entries, we have $b = -b$ and hence $b = 0$. This concludes that

$$A = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$$

is a diagonal matrix. $\square$

5. (a) Let A be a square matrix. Show that $A + A^t$ is a symmetric matrix.

   (b) Let A be an $m \times n$ matrix, show that $A^t A$ is a symmetric matrix.

   Answer.

   (a) $(A + A^t)^t = A^t + (A^t)^t = A^t + A = A + A^t$. Hence $A + A^t$ is symmetric.

   (b) Recall $(AB)^t = B^t A^t$.
   $(A^t A)^t = A^t (A^t)^t = A^t A$. Hence $A^t A$ is symmetric.

$\square$

6. A matrix A is said to be symmetric if $A = A^t$. A matrix A is said to be antisymmetric (also known as skew-symmetric) if $A = -A^t$.
   Obviously A must be a square matrix.
   For a positive integer k, define

$$A^k = \underbrace{A \cdots A}_{k}.$$

   (a) Show that if A is antisymmetric, then A^2 is symmetric.

   (b) Show that if A is antisymmetric, then A^3 is antisymmetric.

   (c) Suppose M is a square matrix with the same size as A. If A is an antisymmetric matrix, Show that MAM^t is also antisymmetric.

   (d) Let k be a positive integer. Show that if A is antisymmetric, then A^k is symmetric (resp. anti-symmetric) if k is even (resp. odd).

(e) Let X be square matrix. Show that X can be written as the sum of a symmetric matrix and an antisymmetric matrix, i.e.

$$X = A + B, \ A \text{ is symmetric and } B \text{ is antisymmetric.}$$

Hint: $A = \frac{X+X^t}{2}, = \frac{X-X^t}{2}$.

(f) Suppose α is a real number, A, B are antisymmetric, then $A + B$ and αA are also antisymmetric.

(g) Let $n = 2$. Prove the following statement:
If A, B are antisymmetric matrices with size n, then AB is symmetric.

(h) Let $n = 3$. Disprove the following statement by finding a counter example:
If A, B are antisymmetric matrices with size n, then AB is symmetric.

Answer.

(a) $(A^2)^t = (AA)^t = A^tA^t = (-A)(-A) = A^2$. Hence A^2 is symmetric.

(b) $(A^3)^t = (AAA)^t = A^tA^tA^t = (-A)(-A)(-A) = (-1)^3A^3 = -A^3$. Hence A^3 is antisymmetric.

(c) $(MAM^t)^t = (M^t)^tA^tM^t = M(-A)M^t = -MAM^t$. Hence MAM^t is antisymmetric.

(d)

$$(A^k)^t = (\underbrace{A \cdots A}_{k})^t = \underbrace{A^t \cdots A^t}_{k} = (-A)^k = (-1)^kA^k.$$

If k is even, then $(A^k)^t = A^k$. Hence A^k is symmetric.
If k is odd, then $(A^k)^t = -A^k$. Hence A^k is antisymmetric.

(e) Let $A = \frac{X+X^t}{2}$, $B = \frac{X-X^t}{2}$.
Then

$$A + B = \frac{X + X^t}{2} + \frac{X - X^t}{2} = \frac{X + X^t + X - X^t}{2} = \frac{2X}{2} = X.$$

Next

$$A^t = \frac{X^t + X^{tt}}{2} = \frac{X^t + X}{2} = A.$$

$$B^t = \frac{X^t - X^{tt}}{2} = -\frac{X - X^t}{2} = -B.$$

Hence A is symmetric and B is antisymmetric.

(f) Because A, B are antisymmetric, $A^t = -A, B^t = -B$.

$$(A + B)^t = A^t + B^t = -A - B = -(A + B).$$

$$(\alpha A)^t = \alpha A^t = -\alpha A.$$

These conclude the results.

(g) Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is antisymmetric. Then

$$A^t = -A.$$

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}.$$

Hence $a = -a$, $b = -c$, $c = -b$ and $d = -d$. Therefore

$$A = \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} = b \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Let $X = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $\alpha = b$. Then $A = \alpha X$. Similarly $B = \beta X$ for some real number β. Then $AB = \alpha\beta X^2 = -\alpha\beta I_2$ is symmetric and hence prove the statement.

(h) Let $A = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$. Both A and B are antisymmetric. But

$$AB = \begin{bmatrix} -1 & -1 & 1 \\ 0 & -2 & 0 \\ 1 & -1 & -1 \end{bmatrix}$$

is not symmetric.

$\square$

7. Let A be a $m \times n$ matrix. Let B be the RREF of A. Let r be the rank of B. Let $\mathbf{b}$ be a column vector of size m. Show that $\mathcal{LS}(A, \mathbf{b})$ is consistent for any $\mathbf{b}$ if and only if $r = m$.

Answer.

As B is the RREF of A, we have a sequence of row operations O_i, such that

$$A \xrightarrow{O_1} \xrightarrow{O_2} \xrightarrow{\cdots} \xrightarrow{O_n} B.$$

And hence we have an sequence of inverse row operations O_i', such that

$$B \xrightarrow{O_n'} \xrightarrow{O_{n-1}'} \xrightarrow{\cdots} \xrightarrow{O_1'} A.$$

(if part: If $r = m$, then $\mathcal{LS}(A, \mathbf{b})$ is consistent for any $\mathbf{b}$.)

Assume $r = m$.
Let $\mathbf{b}$ be arbitrary vector. The system $\mathcal{LS}(A, \mathbf{b})$ have the augmented matrix $[A|\mathbf{b}]$, after the row operations O_i:

$$[A|\mathbf{b}] \xrightarrow{O_1} \xrightarrow{O_2} \xrightarrow{\cdots} \xrightarrow{O_n} [B|\mathbf{b}'], \text{for some vector } \mathbf{b}'.$$

As $r = m$, there is m pivot columns in B, then $[B|\mathbf{b}']$ is already in RREF, and the last column, $\mathbf{b}'$, cannot be pivot. The system $\mathcal{LS}(B,\, \mathbf{b}')$ is consistent. Hence $\mathcal{LS}(A,\, \mathbf{b})$ is also consistent.

(only if part: If $\mathcal{LS}(A,\, \mathbf{b})$ is consistent for any $\mathbf{b}$, then $r = m$.)

Assume $\mathcal{LS}(A,\, \mathbf{b})$ is consistent for any $\mathbf{b}$.
Consider the augmented matrix $[B|\mathbf{e}_m]$, where $\mathbf{e}_m$ is the vector with all entry 0 except the last entry (m-th entry) being 1.
Apply the sequence of row operations O'_i on $[B|\mathbf{e}_m]$, which is:

$$[B|\mathbf{e}_m] \xrightarrow{O'_n} \xrightarrow{O'_{n-1}} \xrightarrow{\cdots} \xrightarrow{O'_1} [A|\mathbf{b_m}], \text{for some vector } \mathbf{b_m}.$$

By the assumption ($\mathcal{LS}(A,\, \mathbf{b})$ is consistent for any $\mathbf{b}$), so the system $\mathcal{LS}(A,\, \mathbf{b}_m)$ is consistent. The system $\mathcal{LS}(B,\, \mathbf{e}_m)$ before row operations is hence also consistent.
As $\mathcal{LS}(B,\, \mathbf{e}_m)$ consistent, last equation will not be $0 = 1$. As last row of B non-zero, B have m pivot column, which is $r = m$.

$\square$

8. Let A be a square matrix of size n such that $AB = BA$ for any square matrix B of size n. Show that $A = cI_n$ for some real number c.
Hint: Consider $B = E_{ij}$, which is a square matrix with all entries being zero except the (i, j)-th being one.

Answer. First, for each $1 \leq i \leq n$, let $B = E_{ii}$.

$$AE_{ii} = \begin{bmatrix} 0 & \cdots & 0 & a_{1i} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & a_{2i} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \\ 0 & \cdots & 0 & a_{ni} & 0 & \cdots & 0 \end{bmatrix}$$

$$E_{ii}A = \begin{bmatrix} 0 & 0 & \vdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \vdots & 0 \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Then $AE_{ii} = E_iA$ implies that $a_{ij} = a_{ji} = 0$ for $j \neq i$. Therefore A is diagonal:

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Next, for each pair of distinct integers $i \neq j$, let $B = E_{ij}$.

$$AE_{ij} = \begin{array}{cc} \begin{array}{ccccccc} & & & j & & & \end{array} & \\ \begin{bmatrix} 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & a_{ii} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{bmatrix} & i \end{array}$$

$$E_{ij}A = \begin{array}{cc} \begin{array}{ccccccc} & & & j & & & \end{array} & \\ \begin{bmatrix} 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & a_{jj} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{bmatrix} & i \end{array}$$

Then $AE_{ij} = E_{ij}A$ implies that $a_{ii} = a_{jj}$. Hence above all, $A = cI_n$ where $c = a_{11}$. $\square$

9. Let A be a 3×5 matrix and

$$A_1 \xrightarrow{2R_1+R_3} A_2 \xrightarrow{R_2 \leftrightarrow R_3} A_3 \xrightarrow{3R_1} A_4.$$

(a) Find a matrix E_1, such that $A_2 = E_1A_1$.

(b) Find a matrix E_2, such that $A_3 = E_2A_2$.

(c) Find a matrix E_3, such that $A_4 = E_3A_3$.

(d) Find a matrix J such that $A_4 = JA_1$. Express J in terms of E_1, E_2, E_3. **You don't need to compute it explicitly**.

(e) Recall the following fact:
Theorem Let A be an $m \times n$ matrix, ρ_i be elementary row operations, for $i = 1, \ldots, k$. Also

$$A \xrightarrow{\rho_1} A_1 \xrightarrow{\rho_2} A_2 \xrightarrow{\rho_2} \cdots \xrightarrow{\rho_k} A_k = B.$$

Define J by applying the same sequence of row operations $\rho_1, \ldots, \rho_k$ on I_m, i.e., J is defined by

$$I_m \xrightarrow{\rho_1} \xrightarrow{\rho_2} \cdots \xrightarrow{\rho_k} J.$$

Then $B = JA$.

Using the theorem, find J defined in the previous part explicitly.

Answer.

(a) E_1 can be obtained by applying the same row operation on I_3:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{2R_1+R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} = E_1.$$

(b) E_2 can be obtained by applying the same row operation on I_3:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = E_2.$$

(c) E_3 can be obtained by applying the same row operation on I_3:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{3R_1} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_3.$$

(d) $A_4 = E_3A_3 = E_3E_2A_2 = E_3E_2E_1A_1$. We can take $J = E_3E_2E_1$.

(e) By the result, J can be obtained by applying the same row operations on I_3:

$$I_3 \xrightarrow{2R_1+R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{3R_1} \begin{bmatrix} 3 & 0 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = J$$

□

10. Let A, B be matrices with four rows. Suppose A and B are row equivalent through the following sequence of row operations.

$$A \xrightarrow{2R_1+R_3} \xrightarrow{3R_2} \xrightarrow{R_1 \leftrightarrow R_4} \xrightarrow{-2R_2+R_3} B.$$

(a) Using the theorem in Question 9e, find a 4×4 matrix G such that $B = GA$.

(b) Find a sequence of elementary row operations that can transform B to A.
Hint: Recall the following facts in the lecture notes

i. If $S_1 \to S_2$ by equation $i \leftrightarrow$ equation j, then S_2 can be transformed to S_1 by equation $i \leftrightarrow$ equation j,

ii. If $S_1 \to S_2$ by $\alpha\times$ equation i, $\alpha \neq 0$. then S_2 can be transformed to S_1 by $\frac{1}{\alpha}\times$ equation i.

iii. If $S_1 \to S_2$ by $\alpha\times$equation i + equation j, then S_2 can be transformed to S_1 by $-\alpha\times$ equation i + equation j,

(c) From the previous part, find a 4×4 matrix H such that $HB = A$.

(d) Compute HG.

Answer.

(a)

$$I_4 \xrightarrow{2R_1+R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{3R_2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1 \leftrightarrow R_4} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 3 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{-2R_2+R_3} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 3 & 0 & 0 \\ 2 & -6 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

We can take

$$G = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 3 & 0 & 0 \\ 2 & -6 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

(b)

$$B \xrightarrow{2R_2+R_3} \xrightarrow{R_1 \leftrightarrow R_4} \xrightarrow{\frac{1}{3}R_2} \xrightarrow{-2R_1+R_3} A.$$

(c)

$$I_4 \xrightarrow{2R_2+R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{3}R_2} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{-2R_1+R_3} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 2 & 1 & -2 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

(d) $HG = I_4$.

$\square$

11. Let

$$A = \begin{bmatrix} 6 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix}.$$

(a) The RREF of A is I_4. Find a sequence of row operations transforms A to I_4.

(b) From the above, and using the theorem in Question 9e, find a matrix H such that $HA = I_4$.

(c) Compute AH.

(d) Reversing the row operations from the part(a), find a sequence of row operations that transforms I_4 to A.

(e) From the above, show that A can be written as a product of elementary row matrices.

Answer.

(a)

$$A \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 6 & 0 & 3 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-2R_2+R_1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 6 & 0 & 3 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{3}R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-2R_1+R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{2R_1+R_4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b)

$$I_4 \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-2R_2+R_1} \begin{bmatrix} 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{3}R_3} \begin{bmatrix} 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-2R_1+R_3} \begin{bmatrix} 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{3} & 4 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{2R_1+R_4} \begin{bmatrix} 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{3} & 4 & -2 & 0 \\ 0 & -4 & 2 & 1 \end{bmatrix} = H$$

(c)

$$AH = \begin{bmatrix} 6 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{3} & 4 & -2 & 0 \\ 0 & -4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(d)

$$I_4 \xrightarrow{-2R_1+R_4} \xrightarrow{2R_1+R_3} \xrightarrow{3R_3} \xrightarrow{2R_2+R_1} \xrightarrow{R_1 \leftrightarrow R_3} A$$

(e)

$$I_4 \xrightarrow{-2R_1+R_4} E_1$$

$$I_4 \xrightarrow{2R_1+R_3} E_2$$

$$I_4 \xrightarrow{3R_3} E_3$$

$$I_4 \xrightarrow{2R_2+R_1} E_4$$

$$I_4 \xrightarrow{R_1 \leftrightarrow R_3} E_5$$

$$A = E_5 E_4 E_3 E_2 E_1$$

□

12. Given the following fact (you don't need to prove the fact):
Fact: Let A, B be $m \times n$ matrices and A is row equivalent to B.
Let S_A be the system of linear equations with the coefficient matrix A and the vector of constants being the zero vector. Let S_B be the system of linear equations with the coefficient matrix B and the vector of constants being the zero vector. Then S_A and S_B are equivalent.

Let

$$A = \begin{bmatrix} 2 & 1 & a & 7 & 1 \\ 1 & 1 & b & 4 & 2 \\ 1 & 1 & c & 4 & 3 \end{bmatrix}$$

and A is row equivalent to an RREF B:

$$A \xrightarrow{\text{RREF}} B = \begin{bmatrix} 1 & 0 & 1 & x & 0 \\ 0 & 1 & -1 & y & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) Find a, b, c.

Hint: $\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ is a solution of S_B.

(b) Find x, y.

Hint: Find a solution of S_A in the form of $\begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

Answer.

(a) From the matrix B, we can find that $\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ is a solution of S_B.

Since A and B are equivalent, then $\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ is a solution of S_A.

Thus, we have

$$\begin{aligned} -2+1+a &= 0 \\ -1+1+b &= 0 \\ -1+1+c &= 0 \end{aligned}$$

So, $a = 1, b = c = 0$.

(b) By matrix A, we know that $\begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is a solution of S_A. This is equivalent to solving

$$\begin{aligned} 2x_1 + x_2 + 7 &= 0 \\ x_1 + x_2 + 4 &= 0 \\ x_1 + x_2 + 4 &= 0 \end{aligned}$$

We obtain $x_1 = -3$, $x_2 = -1$.

Since A and B are equivalent, $\begin{bmatrix} -3 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is a solution of S_B.

Thus, we have

$$\begin{aligned} -3 + x &= 0 \\ -1 + y &= 0 \end{aligned}$$

So, $x = 3, y = 1$.

$\square$

13. Find the RREF and the null space of the following matrices and determine if the matrices are nonsingular.

(a)

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 5 & 8 \\ 1 & 4 & 5 \end{bmatrix}.$$

(b)

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 2 & 1 & 1 & 1 \\ 3 & 0 & 1 & 4 \\ 6 & 1 & 2 & 6 \end{bmatrix}.$$

Answer.

(a)

$$[A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 5 & 8 \\ 1 & 4 & 5 \end{bmatrix} \xrightarrow{-2R_1+R_2} \begin{bmatrix} 1 & 3 & 4 \\ 0 & -1 & 0 \\ 1 & 4 & 5 \end{bmatrix} \xrightarrow{-1R_1+R_3} \begin{bmatrix} 1 & 3 & 4 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix} \xrightarrow{-3R_2+R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix} \xrightarrow{1R_2+R_3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{-1R_3+R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-1R_3+R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Also we obtain the RREF of the matrix A is: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Find the null space of the matrix A is equivalent to find the solution set to the homogeneous system $\mathcal{LS}(A, \mathbf{0})$. We row-reduce the augmented matrix by using the same sequence of row operations to obtain

$$[A|\mathbf{0}] = \left[\begin{array}{ccc|c} 1 & 3 & 4 & 0 \\ 2 & 5 & 8 & 0 \\ 1 & 4 & 5 & 0 \end{array}\right] \xrightarrow{-2R_1+R_2} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 4 & 5 & 0 \end{array}\right] \xrightarrow{-1R_1+R_3} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right] \xrightarrow{-3R_2+R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right] \xrightarrow{1R_2+R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{-1R_3+R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{-1R_3+R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

It has no free variable. Hence the system has only the trivial solution. The null space is $\mathcal{N}(A) = \{\mathbf{0}\}$. Thus the matrix A is nonsingular.

(b) Similarly,

$$[B|\mathbf{0}] = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 \\ 3 & 0 & 1 & 4 & 0 \\ 6 & 1 & 2 & 6 & 0 \end{array}\right] \xrightarrow{-2R_1+R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 3 & 0 & 1 & 4 & 0 \\ 6 & 1 & 2 & 6 & 0 \end{array}\right] \xrightarrow{-3R_1+R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 6 & 1 & 2 & 6 & 0 \end{array}\right]$$

$$\xrightarrow{-6R_1+R_4} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \end{array}\right] \xrightarrow{-1R_2+R_4} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{-1R_3+R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{-1R_3+R_4} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

The system has 1 free variable x_4, which gives the null space

$$\mathcal{N}(B) = \left\{ x_4 \begin{bmatrix} -1 \\ 2 \\ -1 \\ 1 \end{bmatrix} \middle| \; x_4 \in \mathbb{R} \right\}.$$

The RREF of the matrix B is: $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, which implies that the matrix B is singular.

$\square$

14. Let A be a square matrix with size 4 and $\mathbf{b}$ a column vector with size 4. Given that

$$\mathcal{N}(A) = \left\{ x \begin{bmatrix} 1 \\ 2 \\ -1 \\ 4 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix} \middle| \; x, y \in \mathbb{R} \right\}$$

and

$$\mathbf{w} = \begin{bmatrix} 1 \\ 3 \\ 2 \\ -1 \end{bmatrix}$$

is a solution of $\mathcal{LS}(A, \mathbf{b})$. Find the solution set of $\mathcal{LS}(A, \mathbf{b})$.

Answer.

$\mathbf{v}$ is a solution of $\mathcal{LS}(A, \mathbf{b})$ if and only if there exists $x, y \in \mathbb{R}$ such that

$$\mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 2 \\ -1 \end{bmatrix} + x \begin{bmatrix} 1 \\ 2 \\ -1 \\ 4 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, the solution set of $\mathcal{LS}(A, \mathbf{b})$ is given by

$$\left\{ \begin{bmatrix} 1 \\ 3 \\ 2 \\ -1 \end{bmatrix} + x \begin{bmatrix} 1 \\ 2 \\ -1 \\ 4 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix} \middle| \; x, y \in \mathbb{R} \right\}$$

x $\square$

15. Let A be a 3×4 matrix. Given $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \end{bmatrix}$ are solutions of the system of linear equations $\mathcal{LS}(A, \mathbf{b})$.

    (a) Find a nontrivial solution of $\mathcal{LS}(A, \mathbf{0})$.

    (b) Find infinitely many vectors in $\mathcal{N}(A)$.
    Hint: if $\mathbf{x} \in \mathcal{N}(A)$, then $\alpha\mathbf{x}$ is also in $\mathcal{N}(A)$ for any real number α.

    (c) Find infinitely many solutions of $\mathcal{LS}(A, \mathbf{b})$.

    Answer.

    (a) $\mathbf{v} - \mathbf{u}$ is a solution of $\mathcal{LS}(A, \mathbf{0})$, i.e.

    $$\mathbf{v} - \mathbf{u} = \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$$

    is a solution of $\mathcal{LS}(A, \mathbf{0})$.

    (b) The following vectors are in $\mathcal{N}(A)$

    $$\alpha \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \; \alpha \in \mathbf{R}.$$

(c) $\mathbf{v} + \mathbf{z}$ with $\mathbf{z} \in \mathcal{N}(A)$ is a solution of $\mathcal{LS}(A, \mathbf{b})$. From the previous part, the following vectors

$$\begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \quad \alpha \in \mathbb{R}$$

are solutions of $\mathcal{LS}(A, \mathbf{b})$.

$\square$

16. The following sequence of row operations transforms the matrix A to its RREF:

$$A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 1 & 7 \\ 2 & 2 & 15 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 7 \\ 0 & 1 & 3 \\ 2 & 2 & 15 \end{bmatrix} \xrightarrow{-2R_1+R_3} \begin{bmatrix} 1 & 1 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{-1R_2+R_1} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-4R_3+R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-3R_3+R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Let $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Applying the same sequence of row operations to $[A|\mathbf{b}]$, what is the result?

(b) From the previous part, find the solution set of $\mathcal{LS}(A, \mathbf{b})$. Express the answer in terms of b_1, b_2, b_3.

Remark: the method here can be used to show that if A is nonsingular, then $\mathcal{LS}(A, \mathbf{b})$ has a unique solution.

Answer.

(a)

$$[A|\mathbf{b}] = \left[\begin{array}{ccc|c} 0 & 1 & 3 & b_1 \\ 1 & 1 & 7 & b_2 \\ 2 & 2 & 15 & b_3 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 7 & b_2 \\ 0 & 1 & 3 & b_1 \\ 2 & 2 & 15 & b_3 \end{array}\right] \xrightarrow{-2R_1+R_3} \left[\begin{array}{ccc|c} 1 & 1 & 7 & b_2 \\ 0 & 1 & 3 & b_1 \\ 0 & 0 & 1 & b_3 - 2b_2 \end{array}\right]$$

$$\xrightarrow{-1R_2+R_1} \left[\begin{array}{ccc|c} 1 & 0 & 4 & b_2 - b_1 \\ 0 & 1 & 3 & b_1 \\ 0 & 0 & 1 & b_3 - 2b_2 \end{array}\right] \xrightarrow{-4R_3+R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -b_1 + 9b_2 - 4b_3 \\ 0 & 1 & 3 & b_1 \\ 0 & 0 & 1 & b_3 - 2b_2 \end{array}\right]$$

$$\xrightarrow{-3R_3+R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -b_1 + 9b_2 - 4b_3 \\ 0 & 1 & 0 & b_1 + 6b_2 - 3b_3 \\ 0 & 0 & 1 & -2b_2 + b_3 \end{array}\right].$$

(b) The augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -b_1 + 9b_2 - 4b_3 \\ 0 & 1 & 0 & b_1 + 6b_2 - 3b_3 \\ 0 & 0 & 1 & -2b_2 + b_3 \end{array}\right]$$

corresponding to the system of linear equation

$$\begin{aligned} x_1 &= -b_1 + 9b_2 - 4b_3 \\ x_2 &= b_1 + 6b_2 - 3b_3 \\ x_3 &= -2b_2 + b_3 \end{aligned}$$

which gives the unique solution.

□

17. The following sequence of row operations transforms the matrix A to its RREF:

$$A = \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -2 & 0 \\ 3 & 0 & 3 & 10 \\ 2 & 2 & -2 & 8 \end{bmatrix} \xrightarrow{-3R_1+R_3} \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 2 & -2 & 8 \end{bmatrix} \xrightarrow{-2R_1+R_4} \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 2 & -4 & 2 \end{bmatrix}$$

$$\xrightarrow{-2R_2+R_4} \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \xrightarrow{-3R_3+R_1} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \xrightarrow{-2R_3+R_4} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Let $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$.

(a) Applying the same sequence of row operations to $[A|\mathbf{b}]$, what is the result?

(b) From the previous part, find one $\mathbf{b}$ such that $\mathcal{LS}(A, \mathbf{b})$ is inconsistent (i.e. no solution).
Hint: consider the equation corresponding to the last row.

Remark: the method here can be used to show that if A is singular, then $\mathcal{LS}(A, \mathbf{b})$ is inconsistent for some $\mathbf{b}$.

Answer.

(a)

$$[A|\mathbf{b}] = \left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & b_1 \\ 0 & 1 & -2 & 0 & b_2 \\ 3 & 0 & 3 & 10 & b_3 \\ 2 & 2 & -2 & 8 & b_4 \end{array}\right] \xrightarrow{-3R_1+R_3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & b_1 \\ 0 & 1 & -2 & 0 & b_2 \\ 0 & 0 & 0 & 1 & -3b_1 + b_3 \\ 2 & 2 & -2 & 8 & b_4 \end{array}\right]$$

$$\xrightarrow{-2R_1+R_4} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & b_1 \\ 0 & 1 & -2 & 0 & b_2 \\ 0 & 0 & 0 & 1 & -3b_1+b_3 \\ 0 & 2 & -4 & 2 & -2b_1+b_4 \end{array}\right] \xrightarrow{-2R_2+R_4} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & b_1 \\ 0 & 1 & -2 & 0 & b_2 \\ 0 & 0 & 0 & 1 & -3b_1+b_3 \\ 0 & 0 & 0 & 2 & -2b_1-2b_2+b_4 \end{array}\right]$$

$$\xrightarrow{-3R_3+R_1} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 10b_1-3b_3 \\ 0 & 1 & -2 & 0 & b_2 \\ 0 & 0 & 0 & 1 & -3b_1+b_3 \\ 0 & 0 & 0 & 2 & -2b_1-2b_2+b_4 \end{array}\right] \xrightarrow{-2R_3+R_4} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 10b_1-3b_3 \\ 0 & 1 & -2 & 0 & b_2 \\ 0 & 0 & 0 & 1 & -3b_1+b_3 \\ 0 & 0 & 0 & 0 & 4b_1-2b_2-2b_3+b_4 \end{array}\right]$$

(b) The last equation of the system of linear equations with the augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 10b_1-3b_3 \\ 0 & 1 & -2 & 0 & b_2 \\ 0 & 0 & 0 & 1 & -3b_1+b_3 \\ 0 & 0 & 0 & 0 & 4b_1-2b_2-2b_3+b_4 \end{array}\right]$$

is

$$0 = 4b_1 - 2b_2 - 2b_3 + b_4.$$

The system is inconsistent if $4b_1 - 2b_2 - 2b_3 + b_4 \neq 0$. Example, we can take $b_1 = b_2 = b_3 = 0, b_4 = 1$, i.e. $\mathbf{b} = \mathbf{e}_4$.

□

18. For the following matrices, find a nonzero vector in the null space of the given matrices. Justify your answer.

Hint: Recall for an $m \times n$ matrix $A = [\mathbf{A}_1 | \cdots | \mathbf{A}_n]$, where $\mathbf{A}_i$ is the i-th column of A and $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$, then

$$A\mathbf{x} = x_1\mathbf{A}_1 + \cdots + x_n\mathbf{A}_n.$$

(a)

$$A = \begin{bmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & 0 \end{bmatrix}.$$

(b)

$$B = \begin{bmatrix} 1 & 1 & b_{13} \\ 1 & 1 & b_{23} \\ 2 & 2 & b_{33} \\ 3 & 3 & b_{43} \end{bmatrix}.$$

(c)

$$C = \begin{bmatrix} 1 & c_{12} & c_{13} & -2 \\ 2 & c_{22} & c_{23} & -4 \\ 3 & c_{32} & c_{33} & -6 \\ 4 & c_{42} & c_{43} & -8 \end{bmatrix}.$$

Answer.

(a) Let $\mathbf{v} = \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Then

$$A\mathbf{v} = 0 \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} + 0 \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Hence $\mathbf{v} \in \mathcal{N}(A)$.

(b) Let $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Then

$$B\mathbf{u} = 1 \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix} + 0 \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \\ b_{43} \end{bmatrix} = \begin{bmatrix} 1-1 \\ 1-1 \\ 2-2 \\ 3-3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Hence $\mathbf{u} \in \mathcal{N}(B)$.

(c) Let $\mathbf{w} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$. Then

$$C\mathbf{w} = 2 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + 0 \begin{bmatrix} c_{12} \\ c_{22} \\ c_{32} \\ c_{42} \end{bmatrix} + 0 \begin{bmatrix} c_{13} \\ c_{23} \\ c_{33} \\ c_{43} \end{bmatrix} + \begin{bmatrix} -2 \\ -4 \\ -6 \\ -8 \end{bmatrix} = \begin{bmatrix} 2-2 \\ 2\times 2-4 \\ 2\times 3-6 \\ 2\times 4-8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Hence $\mathbf{w} \in \mathcal{N}(C)$.

$\square$

19. Use row operations to transform the following matrix

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 2 & 4 & b \end{bmatrix}$$

in the form

$$\begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & * \end{bmatrix}$$

and hence find b such that the following matrix is singular

Answer.

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 2 & 4 & b \end{bmatrix} \xrightarrow{-1R_1+R_4} \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 2 & 5 & b-1 \end{bmatrix}$$

$$\xrightarrow{-2R_2+R_4} \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & b-3 \end{bmatrix} \xrightarrow{-1R_3+R_4} \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & b-2 \end{bmatrix}$$

$$\xrightarrow{-2R_3+R_2} \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & b-2 \end{bmatrix} \xrightarrow{1R_3+R_1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & b-2 \end{bmatrix}$$

If $b = 2$, The augmented matrix for the homogenous system of linear equations is

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The indexes of the pivot columns are $D = \{1, 2, 3\}$, x_4 is a free variable. The null space is

$$\left\{ \begin{bmatrix} 0 \\ -3x_4 \\ x_4 \\ x_4 \end{bmatrix} \middle| \; x_4 \in \mathbf{R} \right\}.$$

Hence the given matrix is singular for $b = 2$. When $b \neq 2$, then

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & b-2 & 0 \end{array}\right] \xrightarrow{\frac{1}{b-2}R_4} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{1R_4+R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{-3R_4+R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

The indexes of the pivot columns are $D = \{1, 2, 3, 4\}$. There is no free variables. The null space is $\{\mathbf{0}\}$. Hence A is nonsingular for $b \neq 2$.

Therefore, the answer is $b = 2$. $\square$

20. Show that if $\mathbf{u}, \mathbf{v}$ is in $\mathcal{N}(A)$, α a real number. Then $\mathbf{u} + \mathbf{v}$ and $\alpha\mathbf{v}$ are in $\mathcal{N}(A)$.
Hint: $A\mathbf{u} = \mathbf{0}, A\mathbf{v} = \mathbf{0}$.

Answer. Since $\mathbf{u}, \mathbf{v}$ are in $\mathcal{N}(A)$, we have $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$. Then

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

which shows that $\mathbf{u} + \mathbf{v}$ is in $\mathcal{N}(A)$. Similarly,

$$A(\alpha\mathbf{u}) = \alpha(A\mathbf{u}) = \alpha \cdot \mathbf{0} = \mathbf{0}$$

which shows that $\alpha\mathbf{u}$ is in $\mathcal{N}(A)$. $\square$

21. Given $\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ are solutions of the system of linear equations $\mathcal{LS}(A, \mathbf{b})$.

(a) Find a nontrivial solution of $\mathcal{LS}(A, \mathbf{0})$.

(b) Find infinitely many vectors in $\mathcal{N}(A)$.

(c) Find infinitely many solution of $\mathcal{LS}(A, \mathbf{b})$.

Answer. (a) Let $\mathbf{y} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$, then

$$\mathbf{y} - \mathbf{w} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

is in the null space of A.

(b) From (a), $\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$ is in $\mathcal{N}(A)$. By the result of Q20, for any real number α, $\begin{bmatrix} \alpha \\ 3\alpha \\ \alpha \end{bmatrix}$ is in $\mathcal{N}(A)$. Thus we get infinitely many vectors in $\mathcal{N}(A)$.

(c) From (b), $\begin{bmatrix} \alpha \\ 3\alpha \\ \alpha \end{bmatrix}$ is in $\mathcal{N}(A)$ for any real number α. This implies $\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} \alpha \\ 3\alpha \\ \alpha \end{bmatrix}$ is a solution of $\mathcal{LS}(A, \mathbf{b})$ for any real number α. Thus we get infinitely many solutions of $\mathcal{LS}(A, \mathbf{b})$. $\square$

22. Suppose $m > n$ be positive integers. Let A be a $m \times n$ matrix. Let B be a $n \times n$ matrix such that the i-th row of B is the same as the i-th row of A for $i = 1, \ldots, n$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.$$

Show that if the system of equations $\mathcal{LS}(A, \mathbf{0})$ has a non-trivial solution, then B is singular.

Answer. If the system of equations $\mathcal{LS}(A, \mathbf{0})$ has a non-trivial solution, this non-trivial solution must satisfy the first n equations in $\mathcal{LS}(A, \mathbf{0})$ and hence is a non-trivial solution of $\mathcal{LS}(B, \mathbf{0})$. So B is singular. $\square$

23. Let $n \geq 2$ be an integer and A a $n \times n$ square matrix such that the last two columns are identical, show that A is singular.
Hint: Find a non-trivial solution of the system of linear equation $\mathcal{LS}(A, \mathbf{0})$.

Answer. Suppose

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n-1} & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2,n-1} & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{n,n-1} & a_{nn} \end{bmatrix}.$$

with $a_{i,n-1} = a_{in}$ for $i = 1, \ldots, n$. We are going to check that

$$\mathbf{x} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -1 \end{bmatrix} \text{ is in } \mathcal{N}(A),$$

i.e. $\mathbf{x}$ is a nontrivial solution of $\mathcal{LS}(A, \mathbf{0})$:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1,n-1}x_{n-1} + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2,n-1}x_{n-1} + a_{2n}x_n &= 0 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{n,n-1}x_{n-1} + a_{nn}x_n &= 0 \end{aligned}$$

Substitution $x_1 = x_2 = \cdots = x_{n-2} = 0$, $x_{n-1} = 1$, $x_n = -1$ into the left hand of the linear equations. For $i = 1, \ldots, n$, we have

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{i,n-2}x_{n-2} + a_{i,n-1}x_{n-1} + a_{in}x_n$$

$$= a_{i1}0 + a_{i2}0 + \cdots + a_{i,n-2}0 + a_{i,n-1}1 + a_{in}(-1)$$
$$= a_{i,n-1} - a_{in} = 0.$$

Hence $\mathbf{x}$ is a nontrivial solution of the system $\mathcal{LS}(A, \mathbf{0})$. So A is singular.

$\square$

24. Show that if $\mathcal{LS}(A, \mathbf{b})$ is equivalent to $\mathcal{LS}(A', \mathbf{b}')$, and the solution set of $\mathcal{LS}(A, \mathbf{b})$ is non-empty. Then $\mathcal{LS}(A, \mathbf{0})$ is equivalent to $\mathcal{LS}(A', \mathbf{0})$

Answer. First, since the solution set of $\mathcal{LS}(A, \mathbf{b})$ is non-empty, we fix a solution $\mathbf{v_0}$ of $\mathcal{LS}(A, \mathbf{b})$, i.e $A\mathbf{v_0} = \mathbf{b}$. Since $\mathcal{LS}(A, \mathbf{b})$ is equivalent to $\mathcal{LS}(A', \mathbf{b}')$, $\mathbf{v_0}$ is also a solution of $\mathcal{LS}(A', \mathbf{b}')$, i.e $A'\mathbf{v_0} = \mathbf{b}'$.
To show $\mathcal{LS}(A, \mathbf{0})$ is equivalent to $\mathcal{LS}(A', \mathbf{0})$, we have to show that their solution sets are equal, by showing that each solution of $\mathcal{LS}(A, \mathbf{0})$ is a solution of $\mathcal{LS}(A', \mathbf{0})$, and vice versa.
Let $\mathbf{v}$ be a solution of $\mathcal{LS}(A, \mathbf{0})$, i.e. $A\mathbf{v} = \mathbf{0}$. Then

$$A(\mathbf{v} + \mathbf{v}_0) = A\mathbf{v} + A\mathbf{v}_0 = \mathbf{0} + \mathbf{b} = \mathbf{b}$$

which shows that $\mathbf{v} + \mathbf{v}_0$ is a solution of $\mathcal{LS}(A, \mathbf{b})$. Since $\mathcal{LS}(A, \mathbf{b})$ is equivalent to $\mathcal{LS}(A', \mathbf{b}')$, $\mathbf{v} + \mathbf{v}_0$ is also a solution of $\mathcal{LS}(A', \mathbf{b}')$, i.e $A'(\mathbf{v} + \mathbf{v}_0) = \mathbf{b}'$. Note that

$$A'(\mathbf{v} + \mathbf{v}_0) = A'\mathbf{v} + A'\mathbf{v}_0 = A'\mathbf{v} + \mathbf{b}'$$

Combining these, $A'\mathbf{v} + \mathbf{b}' = \mathbf{b}'$, and hence $A'\mathbf{v} = \mathbf{0}$, showing that $\mathbf{v}$ is a solution of $\mathcal{LS}(A', \mathbf{0})$.
Conversely, given a solution $\mathbf{v}'$ of $\mathcal{LS}(A', \mathbf{0})$, we have to show that $\mathbf{v}'$ is a solution of $\mathcal{LS}(A, \mathbf{0})$. This can be done in exactly the same argument as above, by interchanging the roles of A with A', $\mathbf{v}$ with $\mathbf{v}'$, and $\mathbf{b}$ with $\mathbf{b}'$ respectively. $\square$